

Name: _____

Date: _____

INEQUALITIES, (DOUBLE INEQUALITIES)

LO: I can solve and represent double (compound) inequalities.

What is a double inequality?

A **double inequality** has the variable between two bounds, e.g. $3 < x < 7$. It means x is **greater than 3 AND less than or equal to 7** simultaneously.

Strategy: Solve by performing the same operation to all three parts of the inequality.

Quick examples

●	$2 < x < 5$ means x can be 3, 4, or 5 (integers)	inclusive of 5
●	$-1 \leq x < 4$ includes -1 but not 4	closed/open
●	$1 < 2x + 1 < 9$ gives $0 < x < 4$	solve all three parts
●	$2 < x < 5$ means x can be 2	strict inequality excludes 2

Key idea

Whatever you do to the middle, do the same to the left and right. If you multiply/divide by a negative, reverse both inequality signs.

$$a < \boxed{bx + c} \quad d \leq a - \frac{c}{b} < \boxed{x} \quad d - \frac{c}{b}$$

Subtract c , then divide by b (all three parts)

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - listing integers

List the integers satisfying $-2 < x \leq 3$

x must be greater than -2 (not including -2)

x must be less than or equal to 3

Integers: -1, 0, 1, 2, 3

-1, 0, 1, 2, 3

Open circle at -2 (not included), closed at 3 (included).

Example 2 - solving a double inequality

Solve $4 < 2x + 2 \leq 10$

Subtract 2 from all parts: $2 < 2x \leq 8$

Divide all parts by 2: $1 < x \leq 4$

Solution: $1 < x \leq 4$

$1 < x \leq 4$

Perform identical operations on all three parts.

Example 3 - negative coefficient

Solve $-6 \leq -2x < 4$

Divide all parts by -2

Reverse both inequality signs

$3 \leq x < -2$, i.e. $-2 < x \leq 3$

$-2 < x \leq 3$

Dividing by a negative reverses the inequality signs.

Example 4 - harder example

Solve $5 \leq 3x - 1 < 14$

Add 1 to all parts: $6 \leq 3x < 15$

Divide by 3: $2 \leq x < 5$

Solution: $2 \leq x < 5$

$2 \leq x < 5$

Integers satisfying this are 2, 3, 4.

YOUR TURN – PRACTICE (deliberate practice)

1. Listing integers

List all integers satisfying each inequality.

- a) $1 < x \leq 5$
 b) $-3 \leq x < 2$
 c) $0 < x < 4$
 d) $-1 \leq x \leq 3$

2. Solving (add/subtract)

Solve each double inequality.

- a) $3 < x + 1 \leq 7$
 b) $1 \leq x - 2 < 5$
 c) $-4 < x + 3 \leq 6$
 d) $0 \leq x - 5 < 3$

3. Solving (multiply/divide)

Solve each double inequality.

- a) $4 < 2x \leq 10$
 b) $6 \leq 3x < 15$
 c) $2 < \frac{x}{2} \leq 5$
 d) $1 < 4x \leq 20$

4. Two-step solving

Solve and list integer solutions.

- a) $5 < 2x + 1 \leq 11$
 b) $1 \leq 3x - 2 < 10$
 c) $-3 < 2x + 1 \leq 7$
 d) $4 \leq 5x - 1 < 19$

5. Mixed and harder

Solve each inequality fully.

- a) $-6 \leq -3x < 9$ b) $7 < 4x - 1 \leq 19$ c) $-5 \leq 2x + 3 < 11$ d) $0 < x + \frac{1}{2} \leq 3$ e) $-8 < -2(x - 1) \leq 6$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $2 < x \leq 5$ means x can be 3, 4, or 5 (integers) ☐

Correct answer: _____

Reason: _____

b) $2 < x \leq 5$ includes 2 ☐

Correct answer: _____

Reason: _____

c) Dividing by a negative reverses both inequality signs ☐

Correct answer: _____

Reason: _____

d) You only need to solve the right-hand inequality ☐

Correct answer: _____

Reason: _____

e) A double inequality can be split into two separate inequalities ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Find all integers satisfying $-3 < 2x - 1 \leq 9$ AND $x < 4$. Show your working clearly.

Expression: _____

Simplified: _____

b) The perimeter of a rectangle is between **20 cm and 36 cm** (inclusive). The width is **4 cm**. Write a double inequality for the length l and solve it.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
 Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

